IY3660 – Applications of Cryptography

Week 3 Workshop

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Activity 1: Mini-essays

- In the summative assessments for this module I might ask you to write a mini-essay on a specific topic.

- General advice
  1. Recall the definition/contextualise the topic
  2. Answer precisely (compare, justify, argue..)
  3. Stay within the word limit
  4. Use your own words and ideas

Examples – write 250-word essay

- Stream Ciphers vs. Block Ciphers: Which is More Appropriate for Different Applications?

- Cryptographic Strength and Key Length: How Strong is "Strong Enough”

Model answer in bullet points (not the essay)

- Stream Ciphers vs Block Ciphers
  - Performance differences and use cases (e.g., stream ciphers for real-time applications like secure voice communication).
  - Security concerns, such as how key reuse in stream ciphers can lead to vulnerabilities.
  - Real-world examples of each (e.g., RC4 for streams, AES for blocks).
- Discussion might touch several of the following points

- The relationship between key length and computational difficulty (e.g., 128-bit vs. 256-bit AES keys).
- Trade-offs between security and performance when selecting key lengths.
- Guidelines for choosing appropriate key lengths based on use case and current computing capabilities.
- Standard compliance

Activity 2: Example: “matrix” encryption

Imagine a scheme that encrypts n-bit strings P by considering them as a $n\ x\ 1$ column vectors, and multiplying by an invertible secret $n\ x\ n$ matrix M

$$C = M . P$$

Attack strategy (known-plaintext attack) to recover the secret matrix $\boldsymbol{M}$:

- Among the known plaintexts, select any n linear independent vectors P_i .
- Construct a system of n^2 linear equations from the corresponding n ciphertexts C_i
- Solve the system to recover M

WHAT IS THE ATTACK COMPLEXITY?

Namely, how much data is used, how much memory and how much time is needed?

Model answer

Attack complexity:

Data: n plaintext-ciphertext pairs

Memory: ~n^4 bits (to store the n^2 linear equations on n^2 variables)

Time: on the order of n^4 encryption operations (to solve the system of equations)

Activity 3: Quick revision

- Formalise the encryption and decryption steps of one roundo of Feistel cipher.

Model answer

Encryption:

$L_1 = R_0$

$R_1 = L_0 \text{ XOR } f(R_0, K_0)$

Decryption

$R_0 = L_1$

$L_0 = R_1 \text{ XOR } f(L_1, K_0)$